

ASSIGNMENT BRIEF

HTU Course No: 00203360	HTU Course Name: Penetration Testing
BTEC Unit Code:	BTEC UNIT Name:



Student Name/ID Number/Section	
HTU Course Number and Title	00203360 Penetration Testing
BTEC Unit Code and Title	
Academic Year	2025-2026 2
Assignment Author	Sami Almashaqbeh
Course Tutor	Sami Almashaqbeh - Weaam Alrbeiqi
Assignment Title	HtuCrackers
Assignment Ref No	1
Issue Date	29/03/2026
Formative Assessment dates	From 30/03/2026 to 04/06/2026
Submission Date	12/06/2026
IV Name & Date	Elham Derbas 28/03/2026
Submission Format	

The submission for this assignment is:

Two options: the student may choose only one option for submission:

Option 1:

1. OSCP+ from offsec.com, CPTS from HTB , CAPE from HTB, CPENT from EC-council, eCPPT Certification INE Security or any other penetration testing certificate approved by the assessor will cover all Learning outcomes.

Option 2:

1. **An individual written detailed report (in Word format)** that provides thorough, evaluated, or critically reviewed technical information covering all points illustrated in the Assignment Brief and Guidance section. All screenshots must be clear. A signed Student Assessment Submission and Declaration Form must be included in a separate file.
2. **An in-class written exam will be held on 13-06-2026 cover (P1 P4 M1 D1) criteria followed by an practical assessment will be conducted to demonstrate the student's skills. If the student fails to meet any criterion during the practical assessment, that criterion will not be considered fulfilled, even if it was correctly addressed in the submitted report.**

Report guidelines:

In your report, you should make use of headings, paragraphs, and subsections as appropriate. The expected word limit is about 5000 words (recommended 15-20 pages not including snapshots), although you will not be **penalised** for exceeding the total word limit. Do your best to be within the word limit. Your report should be:

1. In a form of a **soft copy** submitted via the URL below.

2. Written in a formal business style using **single spacing and font size 12 Times New Roman (Headings CS)**.

3. Must be supported with research and referenced using the Harvard or APA referencing system.

Note: Soft copies submissions should be done through the university's eLearning system within the deadline specified above from below link: <https://elearning.htu.edu.jo/>

Unit Learning Outcomes

LO1 Gain a solid foundation in penetration testing

LO2 Gain knowledge of CPU instructions, memory addressing, and memory vulnerabilities, while developing advanced exploit development and debugging skills, including mitigation bypass techniques.

LO3 Demonstrate the ability to analyse, secure, and exploit Windows-based systems, Active Directory environments, and advanced web application vulnerabilities using appropriate tools and techniques.

LO4 Demonstrate the ability to locate, analyse, and interpret concealed flags and artefacts within compromised systems in Capture-The-Flag (CTF) exercises.

Assignment Brief and Guidance

HtuCrackers — a mid-sized cybersecurity services firm — invites students to perform an authorized, penetration testing assessment of its simulated environments as part of this course assignment. The company will provide virtual machines and links to several virtual labs (accessible only within the lab network) that replicate its infrastructure. Your objective is to explore the lab links provided, perform discovery and safe vulnerability testing, and locate planted flags or sensitive sample data designed to demonstrate vulnerability impact. Treat the environment as live but isolated: you may attempt controlled exploitation and privilege escalation where necessary to retrieve flags, but you must not attempt real data exfiltration, persistent backdoors, service disruption, or actions outside the provided links, VMs and scope.

Option 1: Penetration testing Certificate:

Submit one of Penetration testing certificate i.e (OSCP+ from offsec.com, CPTS from HTB , CAPE from HTB, CPENT from EC-council, eCPPT Certification INE Security or any other penetration testing certificate approved by the assessor). The certificate will satisfy all Learning outcomes. If you fail to submit the certificate, you will not be considered to have met these criteria, even if you completed most of the certificate content.

Option 2:

Part 1: An in-class written exam will be held on 13-06-2026.

An in-class written exam will cover the following criteria P1 P4 M1 D1.

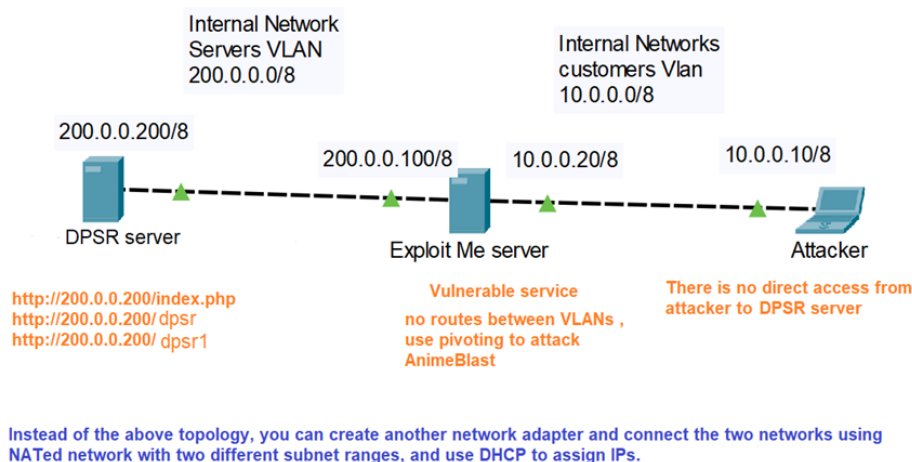
Part 2: Knowledge of CPU instructions, memory addressing, and memory vulnerabilities and developing advanced exploit development and debugging skills:

A: Explain the relationship between the CPU's Instruction Pointer (EIP) and the Stack during a standard function call. How does a stack buffer overflow specifically manipulate this relationship to redirect the 'flow of execution'? Why does a simple buffer overflow often result in application crash, and what specific memory operations must an attacker perform to move from a 'Denial of Service' (crash) to 'Arbitrary Code Execution'?

B: Contrast the reliability of using a JMP ESP instruction versus a Relative Short Jump when bypassing Address Space Layout Randomization (ASLR). In your analysis, explain how the CPU's Stack Pointer (ESP) acts as a dynamic 'bridge' to shellcode, and why a static short jump might fail if the environment's memory map shifts by even a few kilobytes.

C: Target #1 Exploit-Me server: Include screenshots for the main steps

Your setup environment must be as below:



- Download [Exploit-Me server](#) and download [HTUcrackers.zip](#) Use fuzzers such as Spike Fuzzer or any equivalent tool, along with debuggers like Immunity Debugger, to analyze software running in memory and successfully fuzz it to determine ways to crash the software. Hint: use PING command like TRUN.

- Create the full Python script to exploit this service (port 7777) using the memory corruption techniques JMP ESP. Explain each step inside the script using comments, detailing why each line of code is included **using PING as entry point**. Provide proof that your script exploits the target system using a message box for the first exploit, and then using shellcode to create a payload for either a bind shell or a reverse shell for the second exploit. Utilize MSFvenom to generate the necessary payload as Python bytecode.
- Load the backdoor you created through the msfconsole multi-handler, then exploit the target machine
- Create as many tunnels (pivoting points) as needed to reach and attack the [DPSR_server](#). These pivoting points could include FTP, HTTP, and SMB.

Part 3: Ability to analyse, secure, and exploit Windows-based systems, Active Directory environments, and advanced web application vulnerabilities:

In the following tasks, provide only one screenshot as evidence of completion for each task.

A: [portswigger.net Race condition turbo intruder in home assignment](#)

<https://portswigger.net/web-security/learning-paths/race-conditions/race-conditions-detecting-and-exploiting-with-turbo-intruder/race-conditions/lab-race-conditions-bypassing-rate-limits>

B: <https://tryhackme.com/room/operationendgame>

Part 4: Capture The Flag (CTF):

In the following tasks, provide only one screenshot as evidence of completion for each task.

Visit <https://cybertalents.com> and complete the following tasks:

- Five tasks easy level.
- Two tasks medium level.
- Two tasks hard level.

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Gain a solid foundation in penetration testing	P1 Explain the importance of penetration testing in cybersecurity by discussing how it helps assess organizational security, strengthen defences, and outline the various types and strategies used in penetration testing.	M1 Discuss the common areas of penetration testing, the ethical considerations involved, and explain how to initiate, engage, and proceed with a penetration test.	D1 Analyze in detail the risks associated with penetration testing.
LO2 Gain knowledge of CPU instructions, memory addressing, and memory vulnerabilities, while developing advanced exploit development and debugging skills, including mitigation bypass techniques.	P2 Demonstrate understanding of CPU instructions and memory operations, including how vulnerabilities like stack buffer overflows can be used to crash or control software execution. P3 Discuss the differences between static and dynamic code analysis and their role in penetration testing, and using fuzzers and debuggers (e.g., Immunity Debugger) to analyze software in memory and identify vulnerabilities through controlled crashes.	M2 Demonstrate exploit development using memory corruption techniques (e.g., JMP ESP, SEH, ROP) and analyze their effectiveness against defenses such as DEP, ASLR, /GS, and Safe SEH, including common bypass methods.	D2 Analyze CPU pointers and jump strategies (e.g., short, long, forward, back jumps) for exploit control, and implement assembly jump instructions (e.g., via NASM).

LO3 Demonstrate the ability to analyse, secure, and exploit Windows-based systems, Active Directory environments, and advanced web application vulnerabilities using appropriate tools and techniques.	P4 Demonstrate Windows fundamentals and security controls, and explain the fundamental concepts of Active Directory, including the structure of a domain, its core components, and the relationship between forests and domain trusts. P5 Demonstrate, perform, and apply advanced web exploitation techniques, including race conditions, WAF bypass methods, and other complex web attack vectors.	M3 Analyse and demonstrate techniques used to enumerate Active Directory and recover credentials using tools such as Command Prompt, PowerShell, MMC, and BloodHound. M4 Analyse and demonstrate lateral movement and pivoting techniques in a network, such as remote process execution, WMI usage, alternate authentication methods, user behaviour abuse, and port forwarding.	D3 Critically evaluate and exploit advanced Active Directory attack vectors, such as permission delegation, Kerberos delegation, automated relays, user and GPO misconfigurations, certificate services, and domain trust relationships.
LO4 Demonstrate the ability to locate, analyse, and interpret concealed flags and artefacts within compromised systems in Capture-The-Flag (CTF) exercises.	P6 Locate the concealed flags within the compromised system (CTF) (Level 1).	M5 Locate the concealed flags within the compromised system (CTF) (Level 2).	D4 Locate the concealed flags within the compromised system (CTF) (Level 3).